

CS 224

Lab 05

Lab Report

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Section: 04

Report for Matrix Size 1:

Part a) Direct Mapped Caches:

N = 100

Column-by-column:

Cache Size (bytes)	Block Size (words)				
	2	4	8	16	32
128	Hit Rate: 50% Miss Rate: 50% No of Misses: 10824	Hit Rate: 75% Miss Rate: 25% No of Misses: 5413	Hit Rate: 88% Miss Rate: 12% No of Misses: 2707	Hit Rate: 94% Miss Rate: 6% No of Misses: 1354	Hit Rate: 97% Miss Rate: 3% No of Misses: 678
256	Hit Rate: 50% Miss Rate: 50% No of Misses: 10824	Hit Rate: 75% Miss Rate: 25% No of Misses: 5413	Hit Rate: 88% Miss Rate: 12% No of Misses: 2707	Hit Rate: 94% Miss Rate: 6% No of Misses: 1354	Hit Rate: 97% Miss Rate: 3% No of Misses: 678
512	Hit Rate: 50% Miss Rate: 50% No of Misses: 10824	Hit Rate: 75% Miss Rate: 25% No of Misses: 5413	Hit Rate: 88% Miss Rate: 12% No of Misses: 2707	Hit Rate: 94% Miss Rate: 6% No of Misses: 1354	Hit Rate: 97% Miss Rate: 3% No of Misses: 678
1024	Hit Rate: 50% Miss Rate: 50% No of Misses: 10824	Hit Rate: 75% Miss Rate: 25% No of Misses: 5413	Hit Rate: 88% Miss Rate: 12% No of Misses: 2707	Hit Rate: 94% Miss Rate: 6% No of Misses: 1354	Hit Rate: 97% Miss Rate: 3% No of Misses: 678
2048	Hit Rate: 50% Miss Rate: 50% No of Misses: 10824	Hit Rate: 75% Miss Rate: 25% No of Misses: 5413	Hit Rate: 88% Miss Rate: 12% No of Misses: 2707	Hit Rate: 94% Miss Rate: 6% No of Misses: 1354	Hit Rate: 97% Miss Rate: 3% No of Misses: 678

Row-by-row:

Cache Size (bytes)	Block Size (words)				
	2	4	8	16	32
128	Hit Rate: 25% Miss Rate: 75% No of Misses: 15007	Hit Rate: 38% Miss Rate: 62% No of Misses: 12504	Hit Rate: 56% Miss Rate: 44% No of Misses: 11253	Hit Rate: 47% Miss Rate: 53% No of Misses: 10627	Hit Rate: 49% Miss Rate: 51% No of Misses: 10315
256	Hit Rate: 25% Miss Rate:75% No of Misses: 15007	Hit Rate: 38% Miss Rate: 62% No of Misses: 12504	Hit Rate: 56% Miss Rate: 44% No of Misses: 11253	Hit Rate: 47% Miss Rate: 53% No of Misses: 10627	Hit Rate: 49% Miss Rate: 51% No of Misses: 10315
512	Hit Rate: 25% Miss Rate:75% No of Misses: 15007	Hit Rate: 38% Miss Rate: 62% No of Misses: 12504	Hit Rate: 56% Miss Rate: 44% No of Misses: 11253	Hit Rate: 47% Miss Rate: 53% No of Misses: 10627	Hit Rate: 49% Miss Rate: 51% No of Misses: 10315
1024	Hit Rate: 32% Miss Rate: 68% No of Misses: 13607	Hit Rate: 48% Miss Rate: 52% No of Misses: 10404	Hit Rate: 56% Miss Rate: 44% No of Misses: 11253	Hit Rate: 47% Miss Rate: 53% No of Misses: 10627	Hit Rate: 49% Miss Rate: 51% No of Misses: 10315
2048	Hit Rate: 50% Miss Rate: 50% No of Misses: 9997	Hit Rate: 75% Miss Rate: 25% No of Misses: 4999	Hit Rate: 54% Miss Rate: 46% No of Misses: 9153	Hit Rate: 47% Miss Rate: 53% No of Misses: 10627	Hit Rate: 49% Miss Rate: 51% No of Misses: 10315

Part b) Fully Associative Caches:

Row Major:

	Good Hit Rate Cache Size (bytes): 2048 Block Size (words): 4	Medium Hit Rate Cache Size (bytes): 1024 Block Size (words): 8	Low Hit Rate Cache Size (bytes): 128 Block Size (words): 2
Direct Mapped	Hit Rate: 75% Miss Rate: 25% No of Misses: 4999	Hit Rate: 56% Miss Rate: 44% No of Misses: 11253	Hit Rate: 25% Miss Rate: 75% No of Misses: 15007
Fully Associative – LRU	Hit Rate: 75% Miss Rate: 25% No of Misses: 5003	Hit Rate: 45% Miss Rate: 55% No of Misses: 22093	Hit Rate: 25% Miss Rate: 75% No of Misses: 15007
Fully Associative - Random	Hit Rate: 70% Miss Rate: 30% No of Misses: 11942	Hit Rate: 46% Miss Rate: 54% No of Misses: 10841	Hit Rate: 25% Miss Rate: 75% No of Misses: 30005

Part c) N-way Set Associative Caches:

Row Major:

N	Good Hit Rate Cache Size (bytes): 2048 Block Size (words): 4	Medium Hit Rate Cache Size (bytes): 1024 Block Size (words): 8	Low Hit Rate Cache Size (bytes): 128 Block Size (words): 2
4	Hit Rate: 75% Miss Rate: 25% No of Misses: 5003	Hit Rate: 44% Miss Rate: 56% No of Misses: 11253	Hit Rate: 25% Miss Rate: 75% No of Misses: 15007
8	Hit Rate: 75% Miss Rate: 25% No of Misses: 5003	Hit Rate: 44% Miss Rate: 56% No of Misses: 11253	Hit Rate: 25% Miss Rate: 75% No of Misses: 15007
16	Hit Rate: 75% Miss Rate: 25% No of Misses: 5003	Hit Rate: 44% Miss Rate: 56% No of Misses: 11253	Hit Rate: 25% Miss Rate: 75% No of Misses: 15007

Report for Matrix Size 2:

N = 150

Part a) Direct Mapped Caches:

Column-by-column:

Cache Size (bytes)	Block Size (words)				
	2	4	8	16	32
128	Hit Rate: 50% Miss Rate: 50% No of Misses: 23724	Hit Rate: 75% Miss Rate: 25% No of Misses: 11863	Hit Rate: 88% Miss Rate: 12% No of Misses: 5933	Hit Rate: 94% Miss Rate: 6% No of Misses: 2968	Hit Rate: 97% Miss Rate: 3% No of Misses: 1486
256	Hit Rate: 50% Miss Rate: 50% No of Misses: 23724	Hit Rate: 75% Miss Rate: 25% No of Misses: 11863	Hit Rate: 88% Miss Rate: 12% No of Misses: 5933	Hit Rate: 94% Miss Rate: 6% No of Misses: 2968	Hit Rate: 97% Miss Rate: 3% No of Misses: 1486
512	Hit Rate: 50% Miss Rate: 50% No of Misses: 23724	Hit Rate: 75% Miss Rate: 25% No of Misses: 11863	Hit Rate: 88% Miss Rate: 12% No of Misses: 5933	Hit Rate: 94% Miss Rate: 6% No of Misses: 2968	Hit Rate: 97% Miss Rate: 3% No of Misses: 1486
1024	Hit Rate: 50% Miss Rate: 50% No of Misses: 23724	Hit Rate: 75% Miss Rate: 25% No of Misses: 11863	Hit Rate: 88% Miss Rate: 12% No of Misses: 5933	Hit Rate: 94% Miss Rate: 6% No of Misses: 2968	Hit Rate: 97% Miss Rate: 3% No of Misses: 1486
2048	Hit Rate: 50% Miss Rate: 50% No of Misses: 23724	Hit Rate: 75% Miss Rate: 25% No of Misses: 11863	Hit Rate: 88% Miss Rate: 12% No of Misses: 5933	Hit Rate: 94% Miss Rate: 6% No of Misses: 2968	Hit Rate: 97% Miss Rate: 3% No of Misses: 1486

Row-by-row:

Cache Size (bytes)	Block Size (words)				
	2	4	8	16	32
128	Hit Rate: 25%	Hit Rate: 38%	Hit Rate: 44%	Hit Rate: 47%	Hit Rate: 48%

	Miss Rate: 75% No of Misses: 33757	Miss Rate: 62% No of Misses: 28129	Miss Rate: 56% No of Misses: 25316	Miss Rate: 53% No of Misses: 23909	Miss Rate: 52% No of Misses: 23206
256	Hit Rate: 25% Miss Rate: 75% No of Misses: 33757	Hit Rate: 38% Miss Rate: 62% No of Misses: 28129	Hit Rate: 44% Miss Rate: 56% No of Misses: 25316	Hit Rate: 47% Miss Rate: 53% No of Misses: 23909	Hit Rate: 48% Miss Rate: 52% No of Misses: 23206
512	Hit Rate: 25% Miss Rate: 75% No of Misses: 33757	Hit Rate: 38% Miss Rate: 62% No of Misses: 28129	Hit Rate: 44% Miss Rate: 56% No of Misses: 25316	Hit Rate: 47% Miss Rate: 53% No of Misses: 23909	Hit Rate: 48% Miss Rate: 52% No of Misses: 23206
1024	Hit Rate: 43% Miss Rate: 57% No of Misses: 25807	Hit Rate: 42% Miss Rate: 58% No of Misses: 26353	Hit Rate: 44% Miss Rate: 56% No of Misses: 25316	Hit Rate: 47% Miss Rate: 53% No of Misses: 23909	Hit Rate: 48% Miss Rate: 52% No of Misses: 23206
2048	Hit Rate: 50% Miss Rate: 50% No of Misses: 22501	Hit Rate: 58% Miss Rate: 42% No of Misses: 18928	Hit Rate: 46% Miss Rate: 54% No of Misses: 24165	Hit Rate: 47% Miss Rate: 53% No of Misses: 23909	Hit Rate: 48% Miss Rate: 52% No of Misses: 23206

Part b) Fully Associative Caches:

Row Major:

	Good Hit Rate	Medium Hit Rate	Low Hit Rate
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	Cache Size (bytes): 2048 Block Size (words): 4	Cache Size (bytes): 1024 Block Size (words): 8	Cache Size (bytes): 128 Block Size (words): 2
Direct Mapped	Hit Rate: 58% Miss Rate: 42% No of Misses: 18928	Hit Rate: 44% Miss Rate: 56% No of Misses: 25316	Hit Rate: 25% Miss Rate: 75% No of Misses: 33757
Fully Associative – LRU	Hit Rate: 38% Miss Rate: 62% No of Misses: 28129	Hit Rate: 44% Miss Rate: 56% No of Misses: 25316	Hit Rate: 25% Miss Rate: 75% No of Misses: 33757
Fully Associative - Random	Hit Rate: 46% Miss Rate: 54% No of Misses: 48403	Hit Rate: 44% Miss Rate: 56% No of Misses: 50462	Hit Rate: 25% Miss Rate: 75% No of Misses: 67513

Part c) N-way Set Associative Caches:

Row Major:

N	Good Hit Rate Cache Size (bytes): 2048 Block Size (words): 4	Medium Hit Rate Cache Size (bytes): 1024 Block Size (words): 8	Low Hit Rate Cache Size (bytes): 128 Block Size (words): 2
4	Hit Rate: 48% Miss Rate: 52% No of Misses: 23205	Hit Rate: 44% Miss Rate: 56% No of Misses: 25316	Hit Rate: 25% Miss Rate: 75% No of Misses: 33757
8	Hit Rate: 42% Miss Rate: 58% No of Misses: 26025	Hit Rate: 44% Miss Rate: 56% No of Misses: 25316	Hit Rate: 25% Miss Rate: 75% No of Misses: 33757
16	Hit Rate: 38% Miss Rate: 62% No of Misses: 28129	Hit Rate: 44% Miss Rate: 56% No of Misses: 25316	Hit Rate: 25% Miss Rate: 75% No of Misses: 33757

